

AIR UNIVERSITY AEROSPACE AND AVIATION

CAMPUS KAMRA

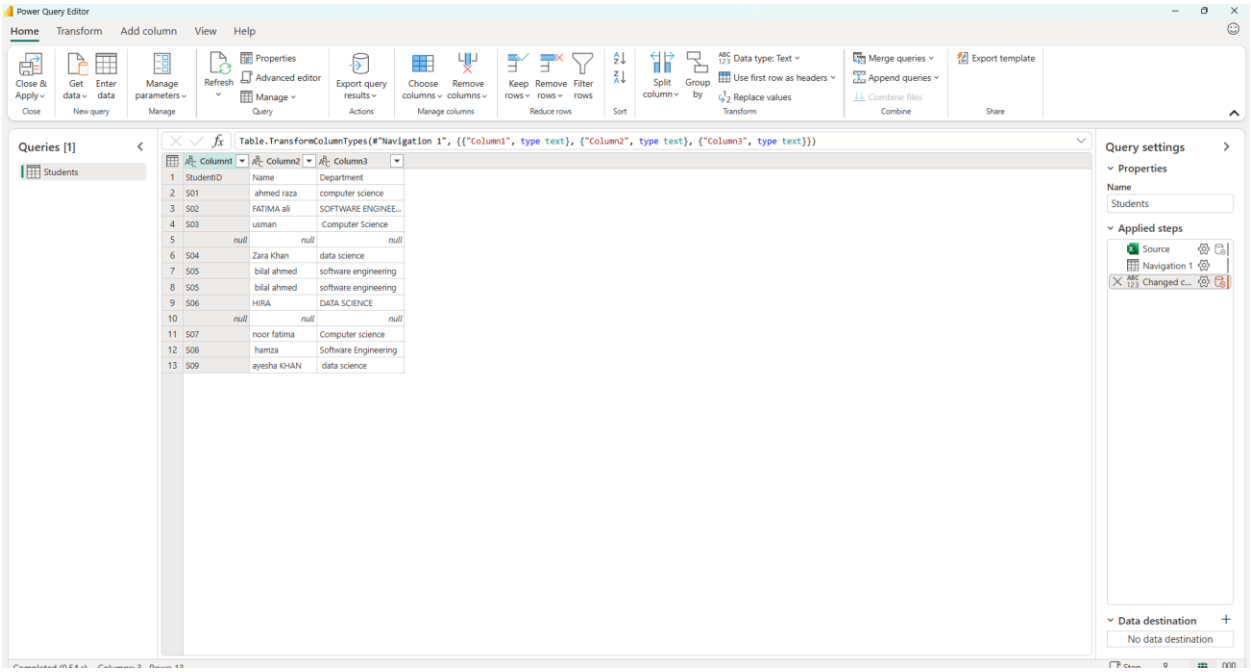
SUBMITTED BY: KHAWAJA MOIN UDDIN

SUBMITTED TO: MR MUSTAJAB HUSSAIN

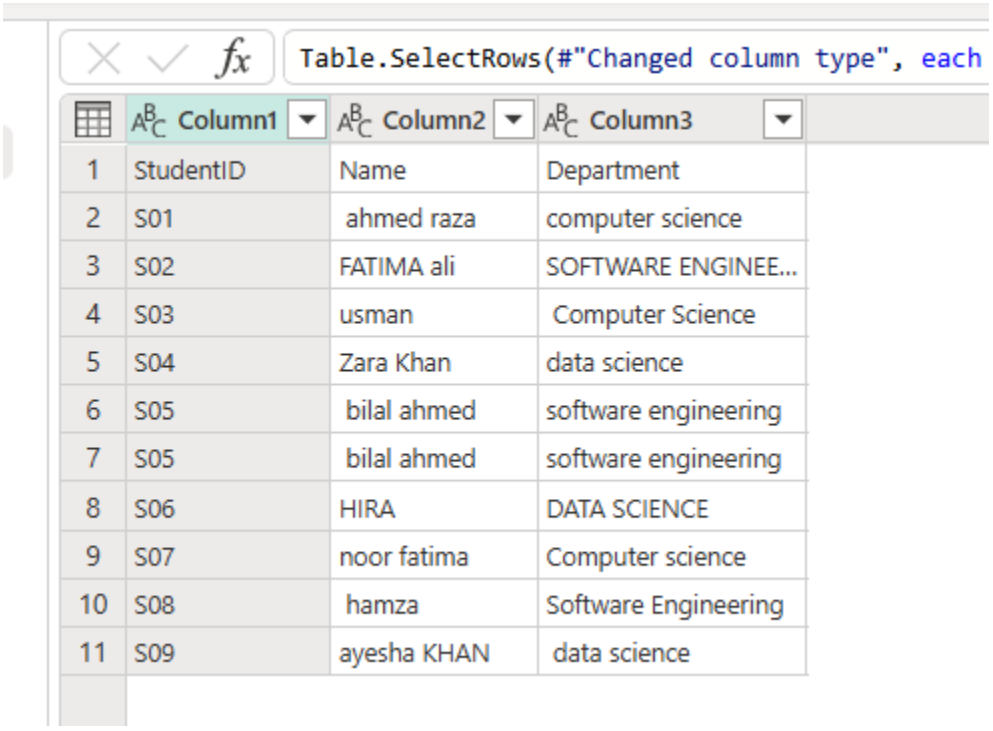
REG ID: 235137

SUBJECT: DS

Task 01:



Remove Blank Rows:



	Column1	Column2	Column3
1	StudentID	Name	Department
2	S01	ahmed raza	computer science
3	S02	FATIMA ali	SOFTWARE ENGINEE...
4	S03	usman	Computer Science
5	S04	Zara Khan	data science
6	S05	bilal ahmed	software engineering
7	S05	bilal ahmed	software engineering
8	S06	HIRA	DATA SCIENCE
9	S07	noor fatima	Computer science
10	S08	hamza	Software Engineering
11	S09	ayesha KHAN	data science

Remove Duplicates:

```
Table.Distinct(#"Promoted headers", {"StudentID"})
```

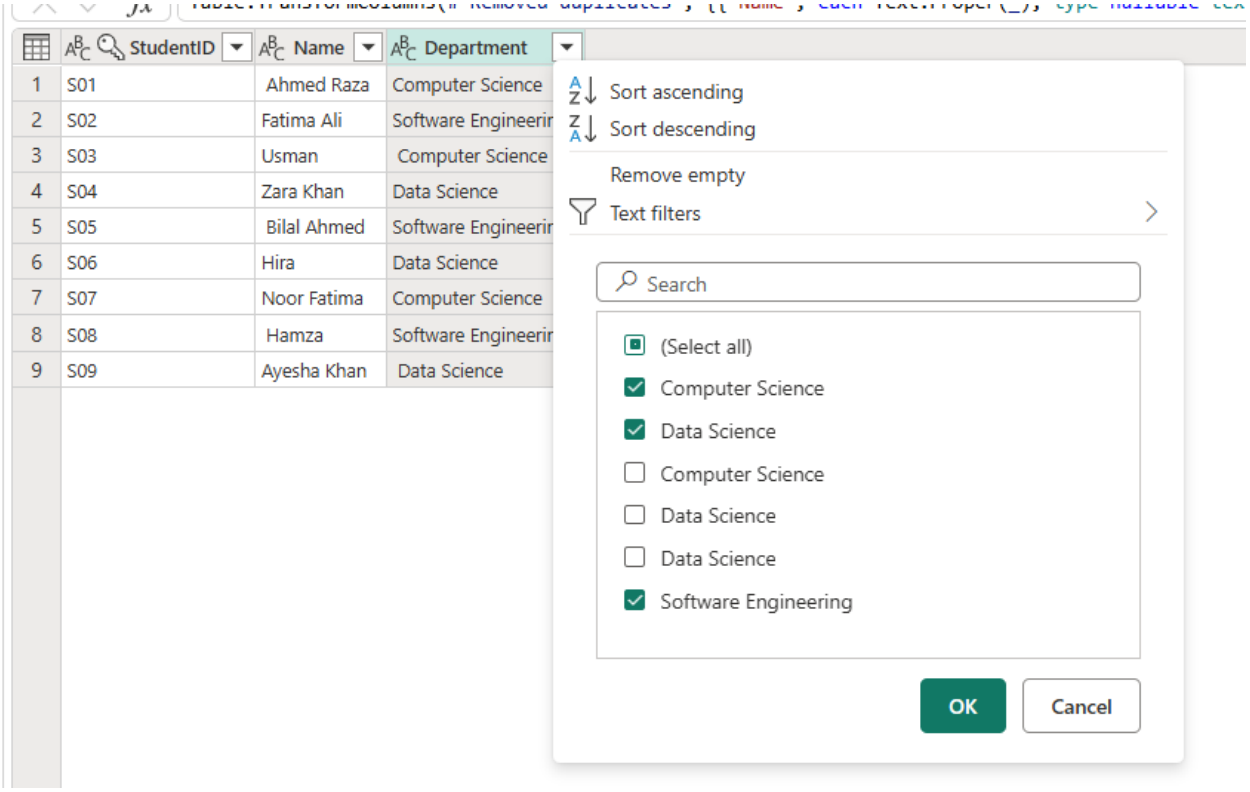
	StudentID	Name	Department
1	S01	ahmed raza	computer science
2	S02	FATIMA ali	SOFTWARE ENGINEE...
3	S03	usman	Computer Science
4	S04	Zara Khan	data science
5	S05	bilal ahmed	software engineering
6	S06	HIRA	DATA SCIENCE
7	S07	noor fatima	Computer science
8	S08	hamza	Software Engineering
9	S09	ayesha KHAN	data science

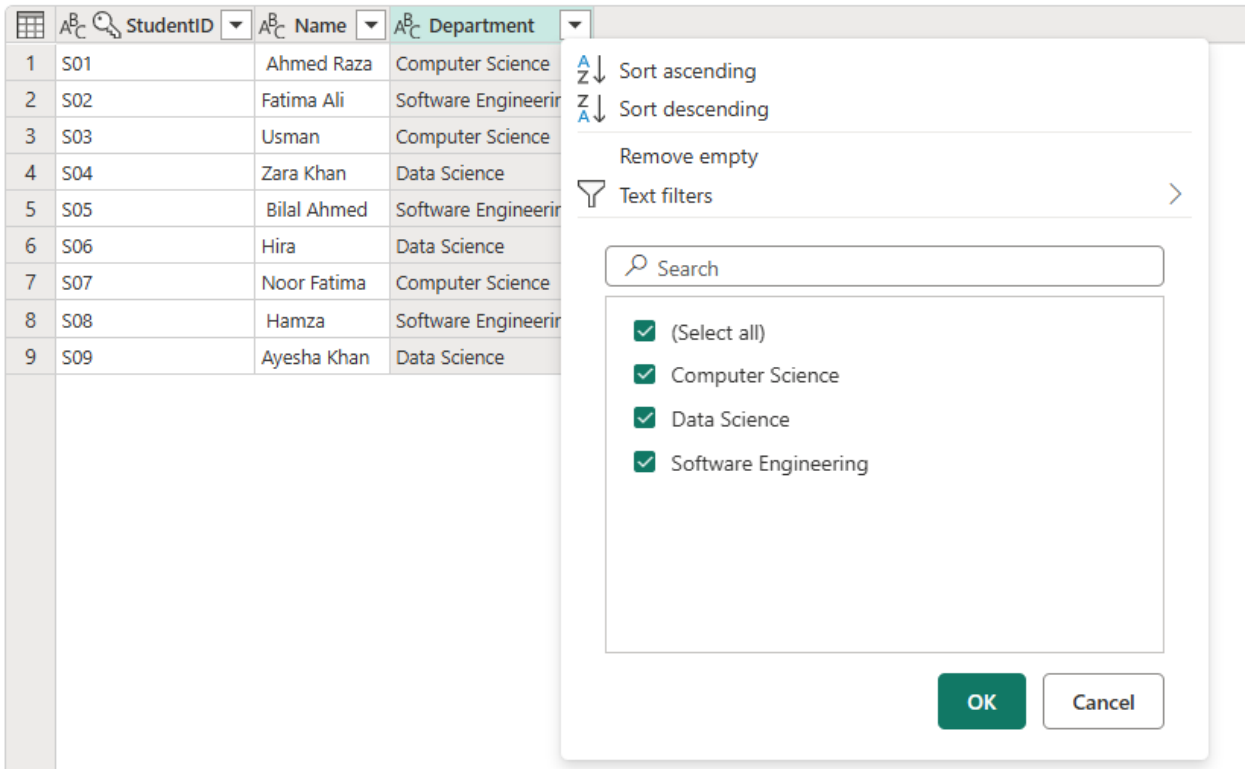
Capitalize Each Word:

```
Table.TransformColumns(#"Removed duplicates"
```

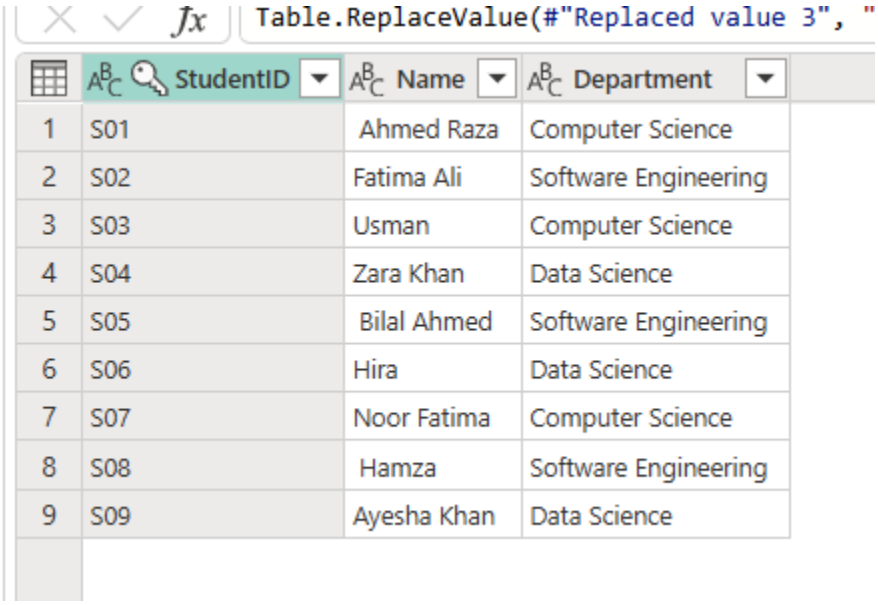
	StudentID	Name	Department
1	S01	Ahmed Raza	Computer Science
2	S02	Fatima Ali	Software Engineering
3	S03	Usman	Computer Science
4	S04	Zara Khan	Data Science
5	S05	Bilal Ahmed	Software Engineering
6	S06	Hira	Data Science
7	S07	Noor Fatima	Computer Science
8	S08	Hamza	Software Engineering
9	S09	Ayesha Khan	Data Science

Only 3 Departments Remain:

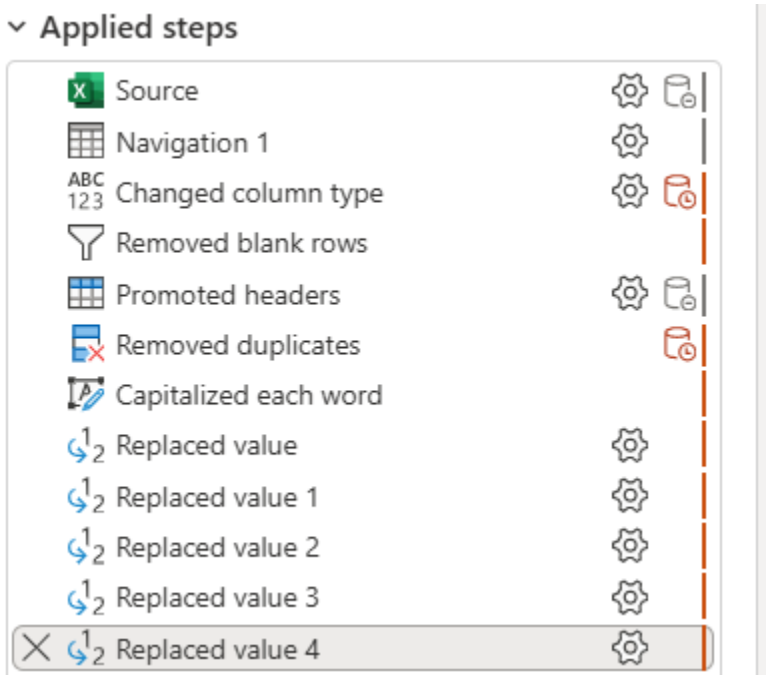


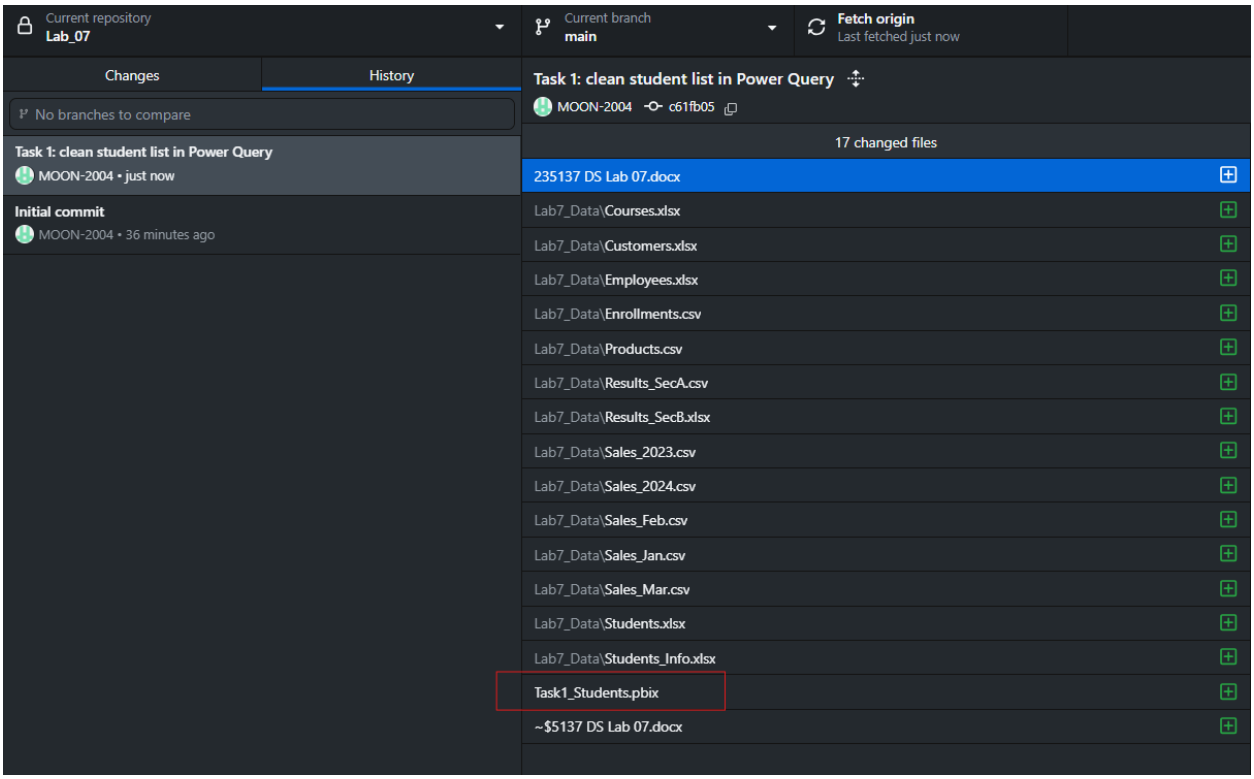


Final Data:

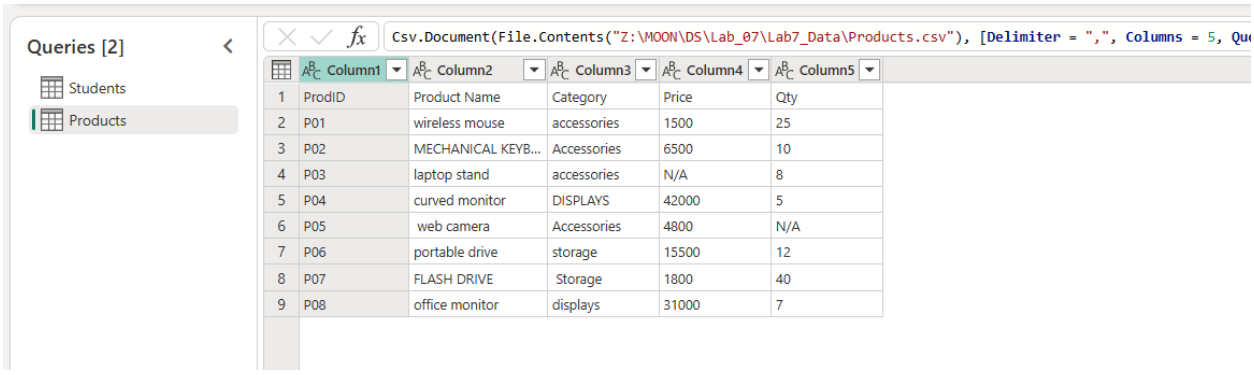


	StudentID	Name	Department
1	S01	Ahmed Raza	Computer Science
2	S02	Fatima Ali	Software Engineering
3	S03	Usman	Computer Science
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6	S06	Hira	Data Science
7	S07	Noor Fatima	Computer Science
8	S08	Hamza	Software Engineering
9	S09	Ayesha Khan	Data Science



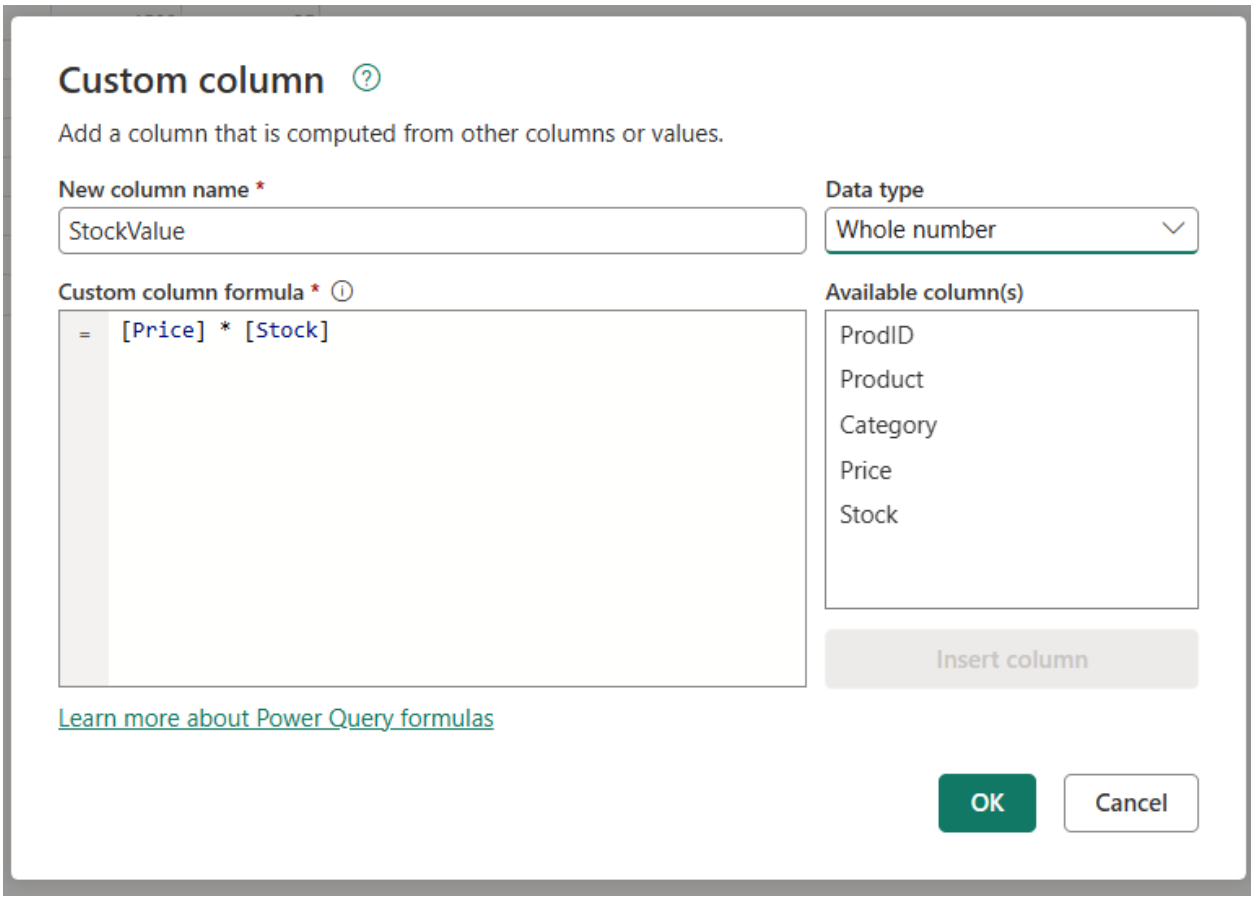


Task 02:

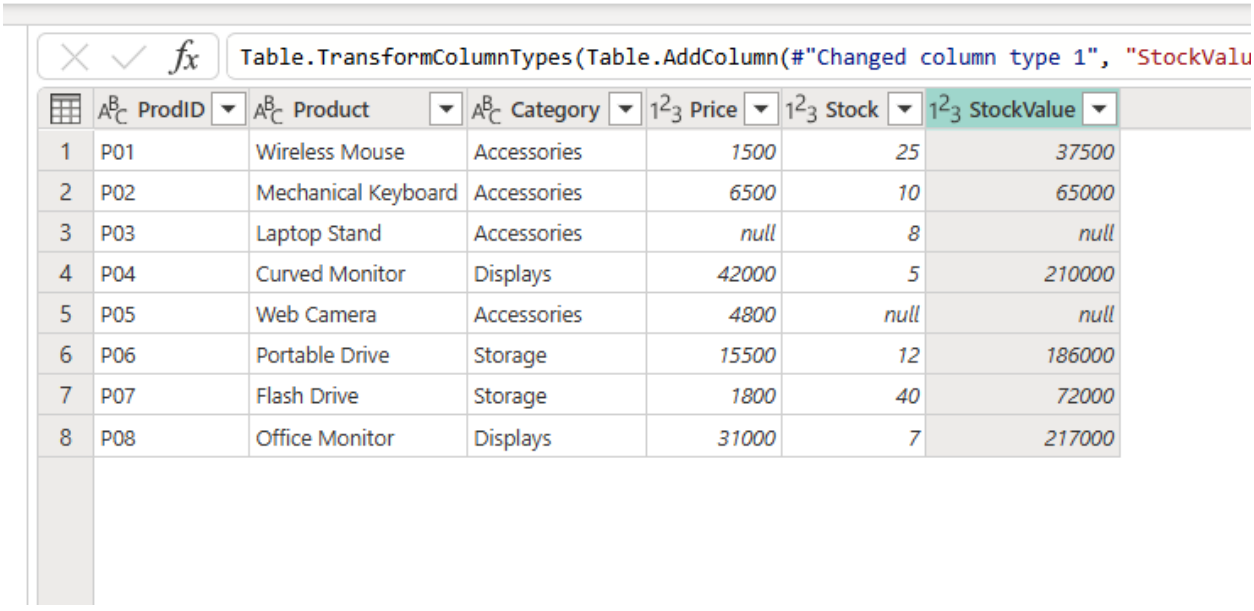


	Column1	Column2	Column3	Column4	Column5
1	ProdID	Product Name	Category	Price	Qty
2	P01	wireless mouse	accessories	1500	25
3	P02	MECHANICAL KEYB...	Accessories	6500	10
4	P03	laptop stand	accessories	N/A	8
5	P04	curved monitor	DISPLAYS	42000	5
6	P05	web camera	Accessories	4800	N/A
7	P06	portable drive	storage	15500	12
8	P07	FLASH DRIVE	Storage	1800	40
9	P08	office monitor	displays	31000	7

Add Custom Column:



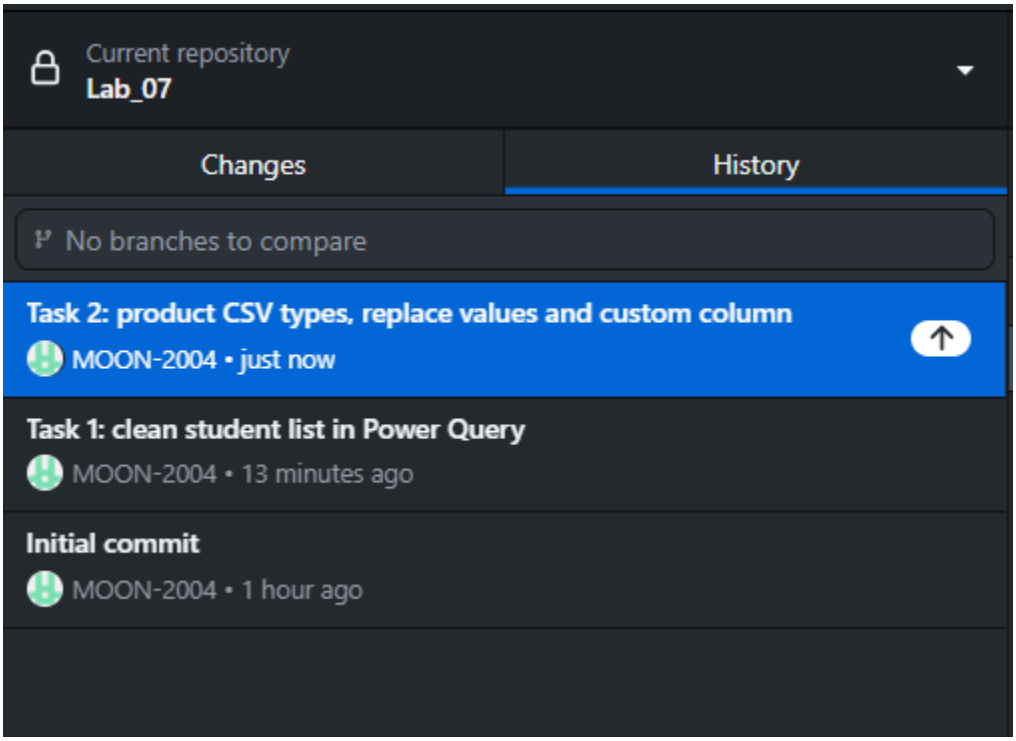
Final Data:



	ProdID	Product	Category	Price	Stock	StockValue
1	P01	Wireless Mouse	Accessories	1500	25	37500
2	P02	Mechanical Keyboard	Accessories	6500	10	65000
3	P03	Laptop Stand	Accessories	null	8	null
4	P04	Curved Monitor	Displays	42000	5	210000
5	P05	Web Camera	Accessories	4800	null	null
6	P06	Portable Drive	Storage	15500	12	186000
7	P07	Flash Drive	Storage	1800	40	72000
8	P08	Office Monitor	Displays	31000	7	217000

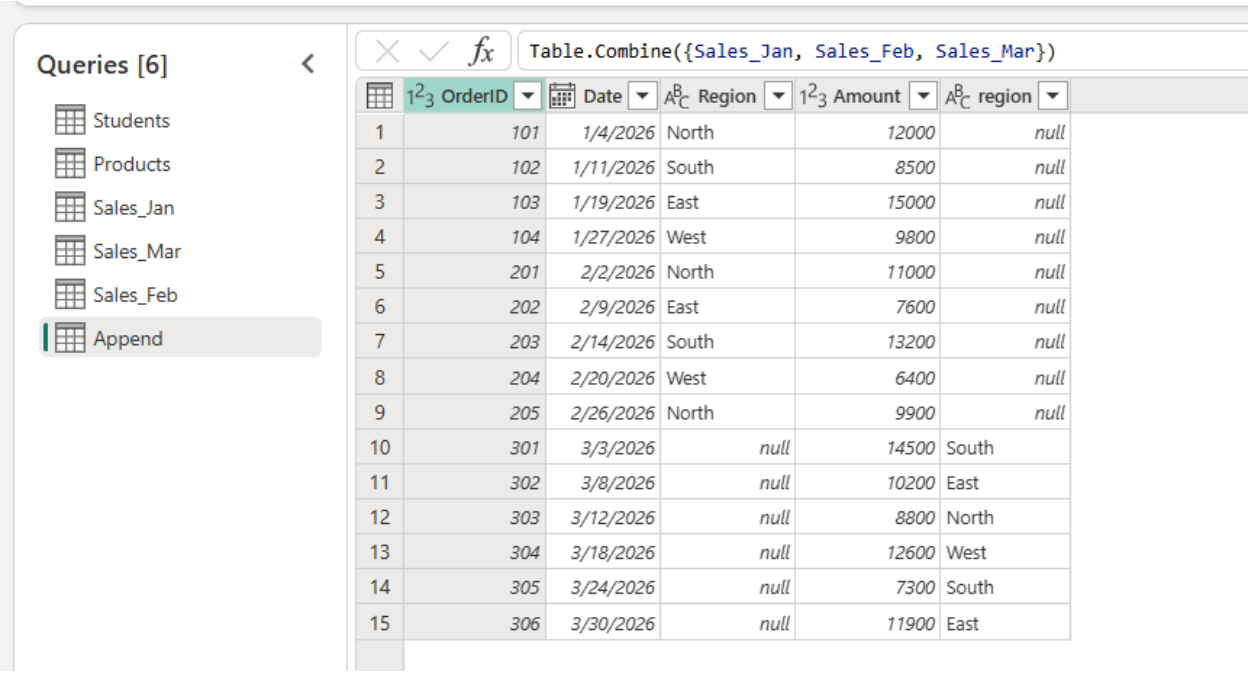
Applied steps

- Source
- Promoted headers
- Replaced value
- Renamed columns
- Capitalized each word
- Trimmed text
- Replaced value 1
- Changed column type
- Replaced value 2
- Changed column type 1
- Added custom



Task 03:

Append Tables:

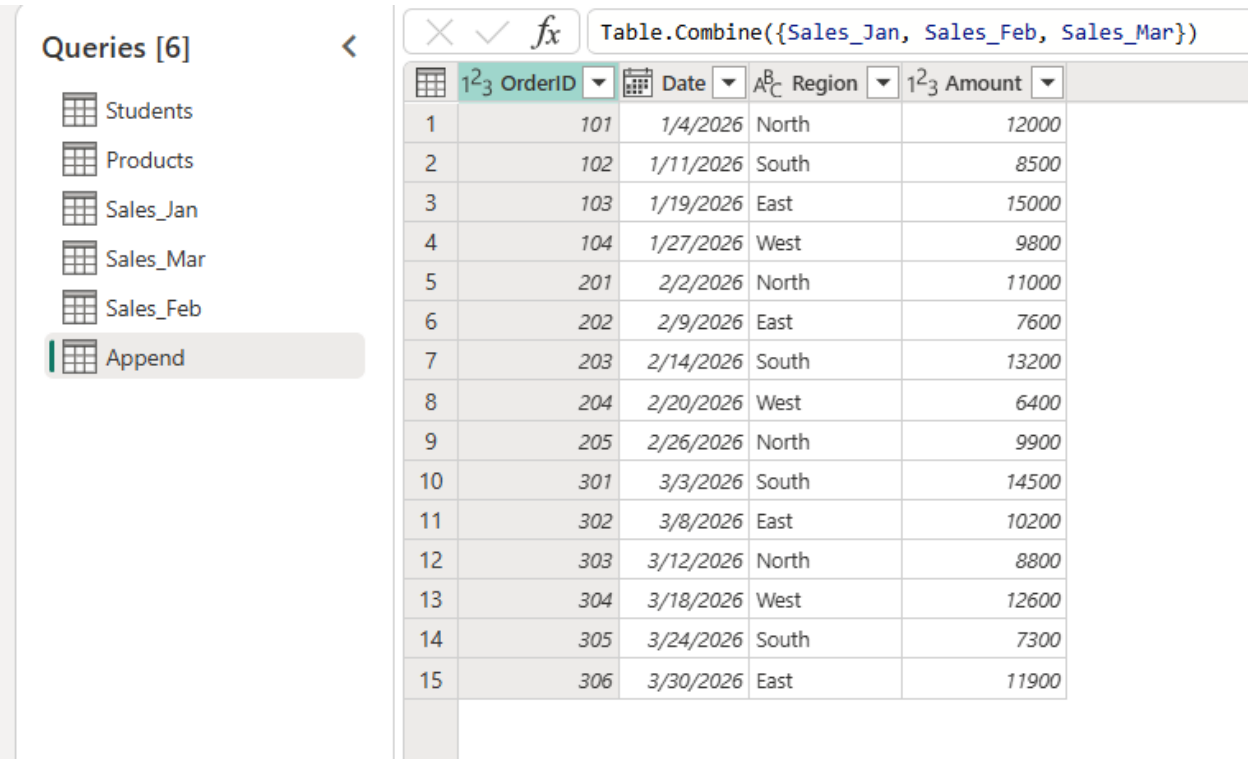


Queries [6]

- Students
- Products
- Sales_Jan
- Sales_Mar
- Sales_Feb
- Append

```
Table.Combine({Sales_Jan, Sales_Feb, Sales_Mar})
```

	OrderID	Date	Region	Amount	region
1	101	1/4/2026	North	12000	null
2	102	1/11/2026	South	8500	null
3	103	1/19/2026	East	15000	null
4	104	1/27/2026	West	9800	null
5	201	2/2/2026	North	11000	null
6	202	2/9/2026	East	7600	null
7	203	2/14/2026	South	13200	null
8	204	2/20/2026	West	6400	null
9	205	2/26/2026	North	9900	null
10	301	3/3/2026	null	14500	South
11	302	3/8/2026	null	10200	East
12	303	3/12/2026	null	8800	North
13	304	3/18/2026	null	12600	West
14	305	3/24/2026	null	7300	South
15	306	3/30/2026	null	11900	East



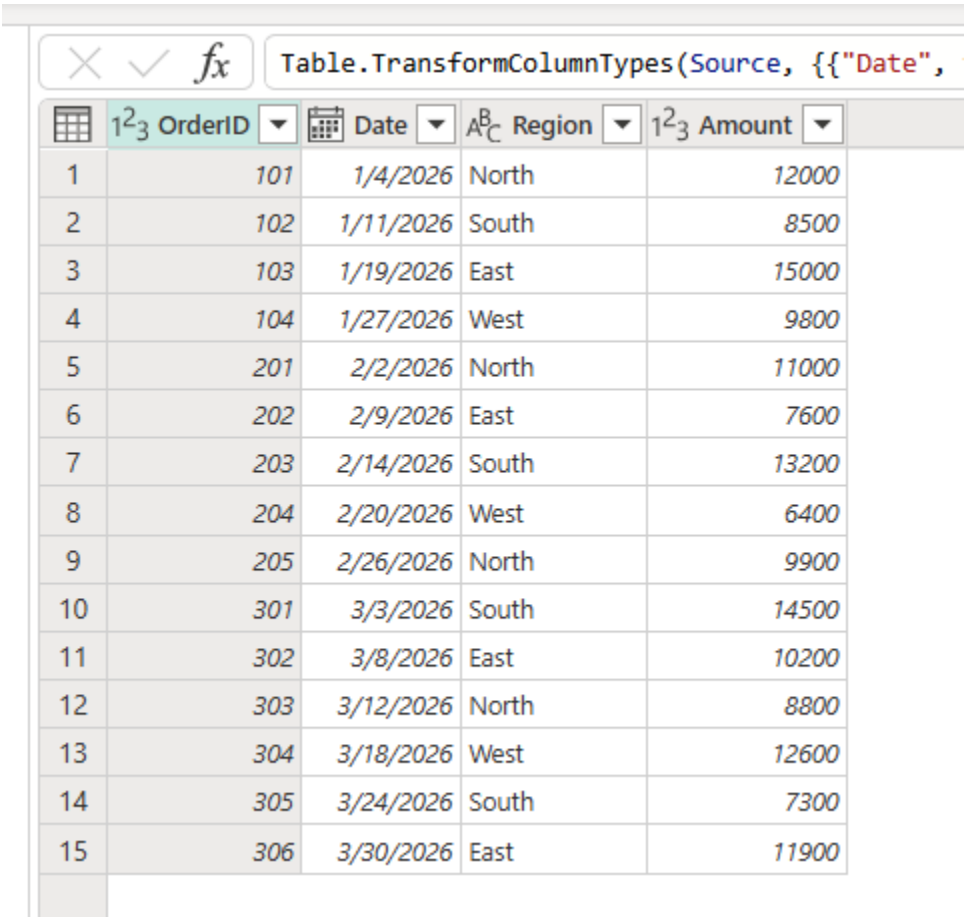
Queries [6]

- Students
- Products
- Sales_Jan
- Sales_Mar
- Sales_Feb
- Append

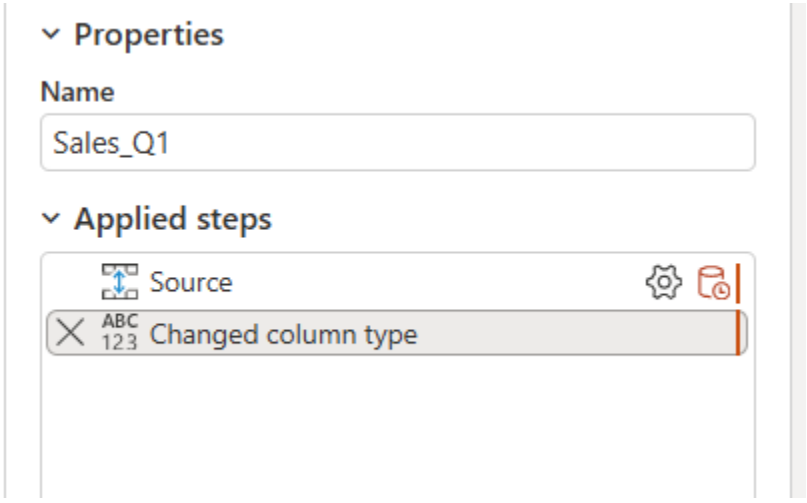
```
Table.Combine({Sales_Jan, Sales_Feb, Sales_Mar})
```

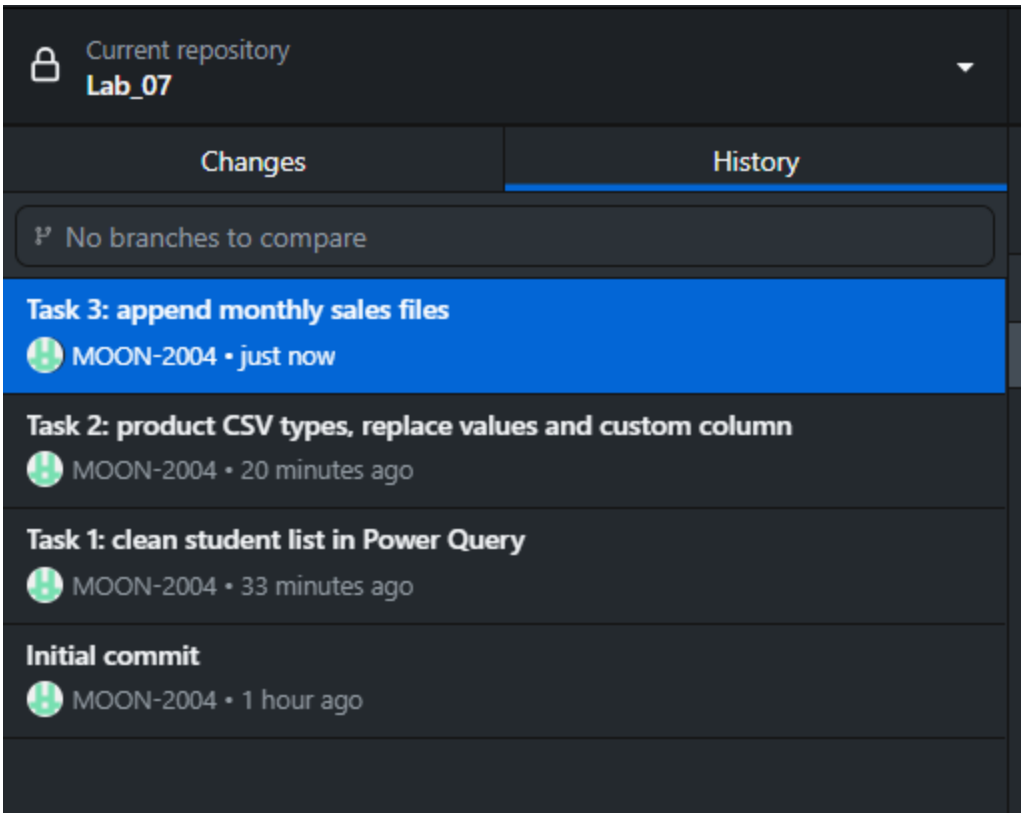
	OrderID	Date	Region	Amount
1	101	1/4/2026	North	12000
2	102	1/11/2026	South	8500
3	103	1/19/2026	East	15000
4	104	1/27/2026	West	9800
5	201	2/2/2026	North	11000
6	202	2/9/2026	East	7600
7	203	2/14/2026	South	13200
8	204	2/20/2026	West	6400
9	205	2/26/2026	North	9900
10	301	3/3/2026	South	14500
11	302	3/8/2026	East	10200
12	303	3/12/2026	North	8800
13	304	3/18/2026	West	12600
14	305	3/24/2026	South	7300
15	306	3/30/2026	East	11900

Final Data:

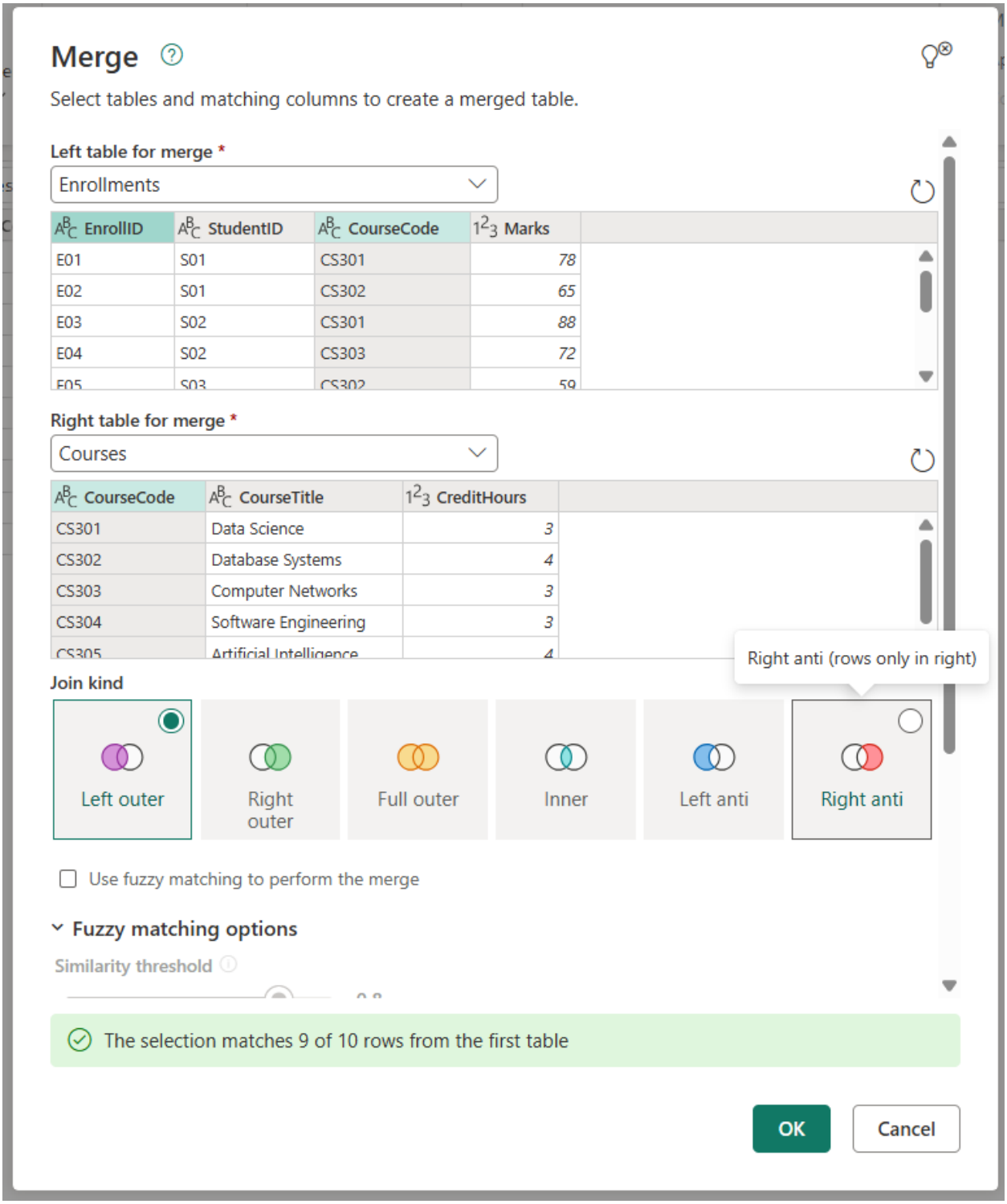


	OrderID	Date	Region	Amount
1	101	1/4/2026	North	12000
2	102	1/11/2026	South	8500
3	103	1/19/2026	East	15000
4	104	1/27/2026	West	9800
5	201	2/2/2026	North	11000
6	202	2/9/2026	East	7600
7	203	2/14/2026	South	13200
8	204	2/20/2026	West	6400
9	205	2/26/2026	North	9900
10	301	3/3/2026	South	14500
11	302	3/8/2026	East	10200
12	303	3/12/2026	North	8800
13	304	3/18/2026	West	12600
14	305	3/24/2026	South	7300
15	306	3/30/2026	East	11900




Current repository
Lab_07
Changes
History
No branches to compare
Task 3: append monthly sales files
MOON-2004 • just now
Task 2: product CSV types, replace values and custom column
MOON-2004 • 20 minutes ago
Task 1: clean student list in Power Query
MOON-2004 • 33 minutes ago
Initial commit
MOON-2004 • 1 hour ago

Task 04:

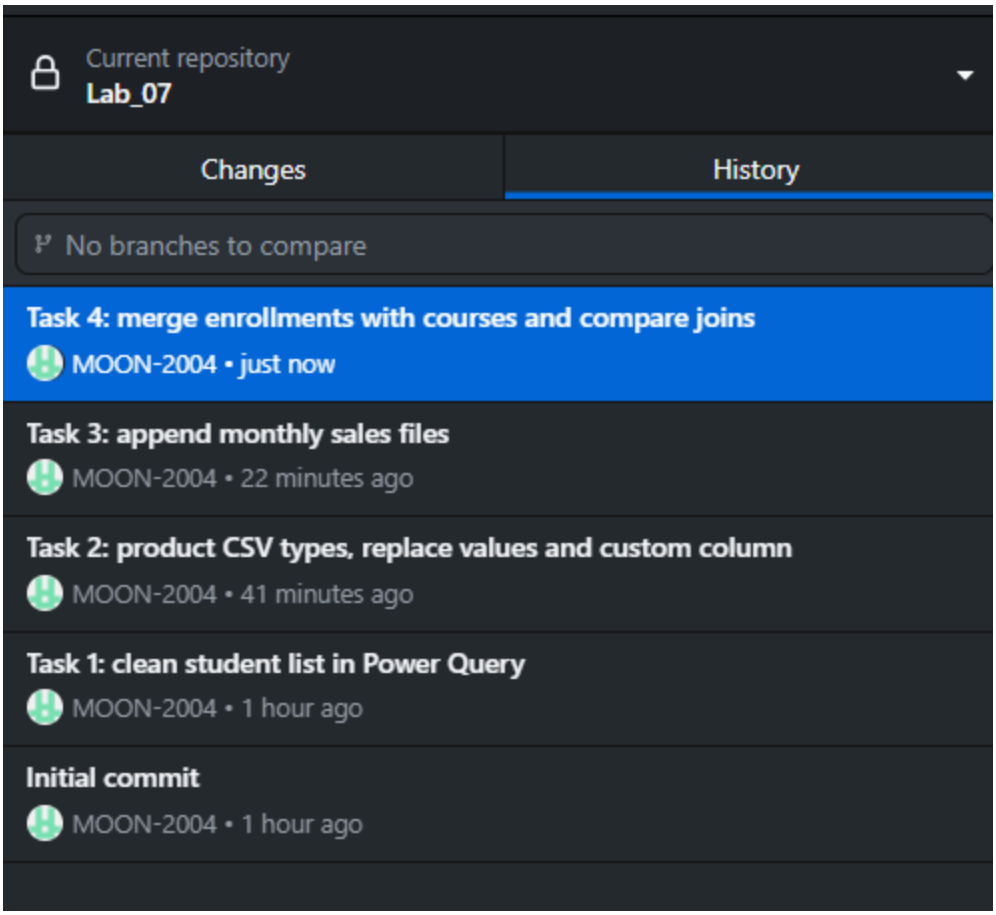

Merge
Select tables and matching columns to create a merged table.
Left table for merge *
Enrollments
EnrollID StudentID CourseCode Marks
E01 S01 CS301 78
E02 S01 CS302 65
E03 S02 CS301 88
E04 S02 CS303 72
E05 S03 CS302 59
Right table for merge *
Courses
CourseCode CourseTitle CreditHours
CS301 Data Science 3
CS302 Database Systems 4
CS303 Computer Networks 3
CS304 Software Engineering 3
CS305 Artificial Intelligence 4
Right anti (rows only in right)
Join kind
Left outer
Right outer
Full outer
Inner
Left anti
Right anti
Use fuzzy matching to perform the merge
Fuzzy matching options
Similarity threshold
The selection matches 9 of 10 rows from the first table
OK
Cancel

```
Rows count:
Left Outer: 10
Inner: 9
Full Outer: 11
Left Anti joins: 1
```

Final Data (Enroll_Details):

```
Table.ExpandTableColumn(Source, "Courses", {"CourseTitle", "CreditHours"}, {"CourseTitle",
```

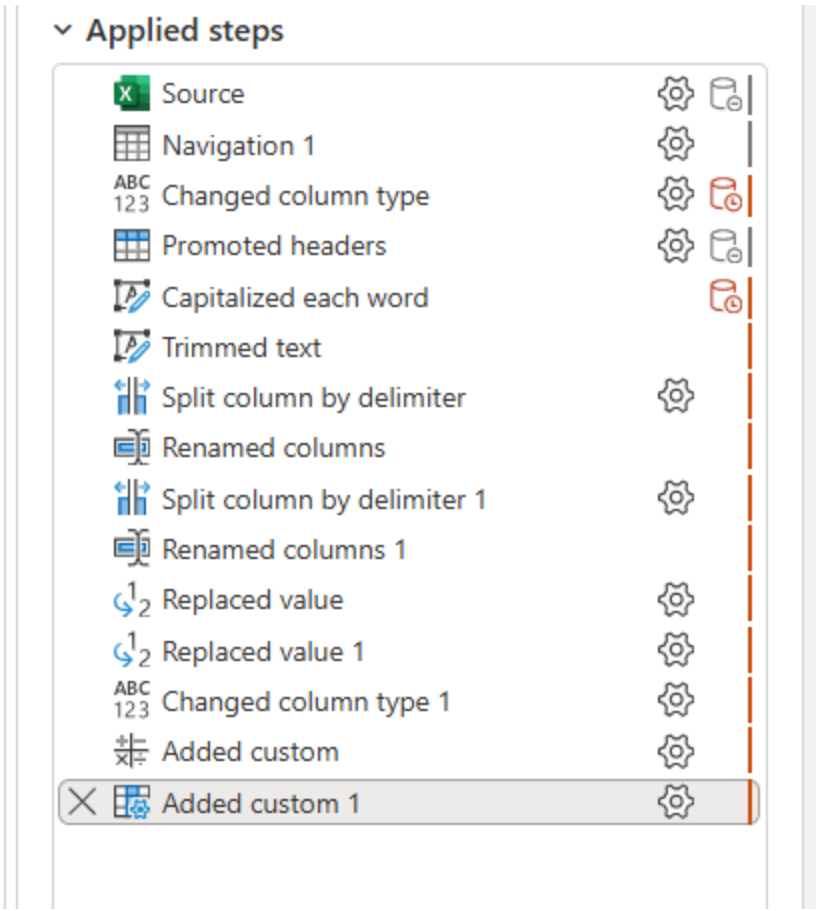
	EnrollID	StudentID	CourseCode	Marks	CourseTitle	CreditHours
1	E01	S01	CS301	78	Data Science	3
2	E03	S02	CS301	88	Data Science	3
3	E02	S01	CS302	65	Database Systems	4
4	E05	S03	CS302	59	Database Systems	4
5	E04	S02	CS303	72	Computer Networks	3
6	E06	S03	CS304	81	Software Engineering	3
7	E07	S04	CS301	91	Data Science	3
8	E08	S04	CS303	67	Computer Networks	3
9	E09	S05	CS304	74	Software Engineering	3
10	E10	S05	CS999	70	null	null

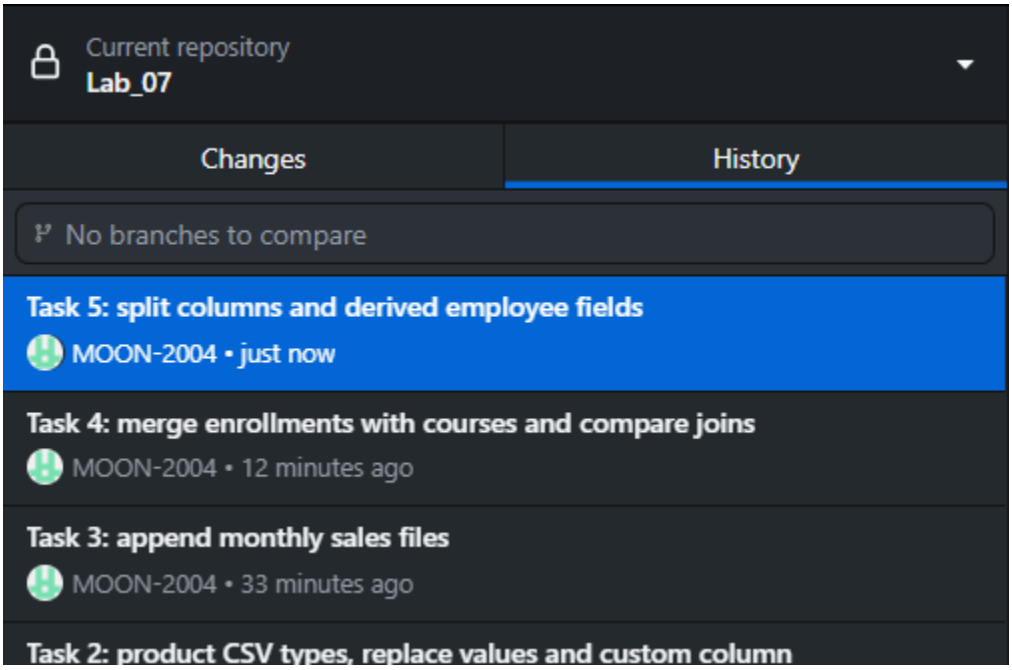


Task 05:

Final Data:

	EmpID	FirstName	LastName	Department	City	Salary	AnnualSalary	Email
1	E01	Ali	Raza	sales	lahore	85000	1020000	ali.raza@company.pk
2	E02	Sana	Malik	MARKETING	islamabad	120000	1440000	sana.malik@company.pk
3	E03	Omar	Farooq	finance	Karachi	95500	1146000	omar.farooq@company.pk
4	E04	Maryam	Iqbal	Sales	Islamabad	110000	1320000	maryam.iqbal@company...
5	E05	Faisal	Nadeem	marketing	lahore	78000	936000	faisal.nadeem@company...
6	E06	Kiran	Shah	Finance	LAHORE	102000	1224000	kiran.shah@company.pk

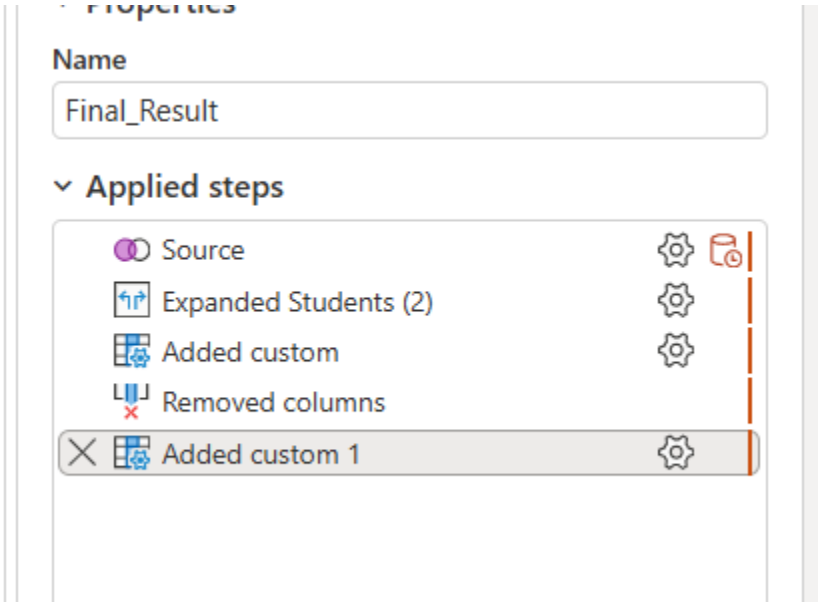

Applied steps
Source
Navigation 1
Changed column type
Promoted headers
Capitalized each word
Trimmed text
Split column by delimiter
Renamed columns
Split column by delimiter 1
Renamed columns 1
Replaced value
Replaced value 1
Changed column type 1
Added custom
Added custom 1

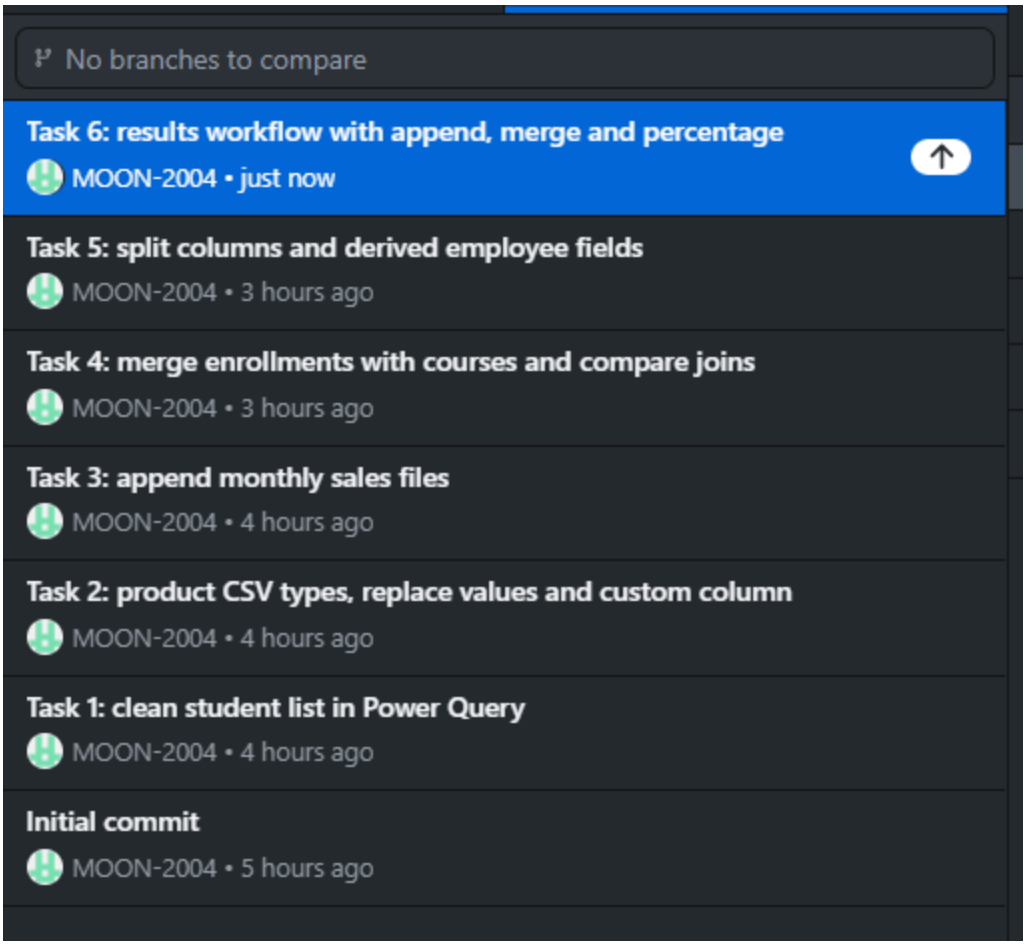

Current repository
Lab_07
Changes
History
No branches to compare
Task 5: split columns and derived employee fields
MOON-2004 • just now
Task 4: merge enrollments with courses and compare joins
MOON-2004 • 12 minutes ago
Task 3: append monthly sales files
MOON-2004 • 33 minutes ago
Task 2: product CSV types, replace values and custom column

Task 06:

Final Result:

	RegNo	Obtained	TotalMarks	Section	Name	Program	Percentage
1	AU101	78	100	A	Ali Hassan	BSCS	78
2	AU102	85	100	A	Sara Khan	BSCS	85
3	AU103	null	100	A	Usman Tariq	BSCS	null
4	AU104	64	100	A	Hina Javed	BSCS	64
5	AU105	91	100	A	Bilal Akram	BSCS	91
6	AU106	36	50	B	Zainab Noor	BSCS	72
7	AU107	44	50	B	Hamza Iqbal	BSCS	88
8	AU108	30	50	B	Mehwish Ali	BSCS	60
9	AU109	34	50	B	Saad Rehman	BSCS	68
10	AU111	40	50	B	null	null	80




No branches to compare
Task 6: results workflow with append, merge and percentage
MOON-2004 • just now
Task 5: split columns and derived employee fields
MOON-2004 • 3 hours ago
Task 4: merge enrollments with courses and compare joins
MOON-2004 • 3 hours ago
Task 3: append monthly sales files
MOON-2004 • 4 hours ago
Task 2: product CSV types, replace values and custom column
MOON-2004 • 4 hours ago
Task 1: clean student list in Power Query
MOON-2004 • 4 hours ago
Initial commit
MOON-2004 • 5 hours ago